A COMPREHENSIVE STUDY ON SOCIAL NETWORK MENTAL DISORDERS DETECTION VIA ONLINE SOCIAL MEDIA MINING

A Mini Project Report

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In partial fulfillment of the requirements for the

Award of the degree of

BACHELOR OF TECHNOLOGY

In

CSE

By

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CERTIFICATE

This is to certify that the Mini Project Report on ***"A COMPREHENSIVE STUDY ON SOCIAL NETWORK MENTAL DISORDERS DETECTION VIA ONLINE SOCIAL MEDIA MINING"*** submitted by **NAGULA SOWMYA** bearing Hall ticket No. **17VE1A05F6** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **COMPUTER SCIENCE AND ENGINEERING** from Jawaharlal Nehru Technological University, Kukatpally, Hyderabad for the academic year 2020-21 is a record of bonafide work carried out by them under our guidance and Supervision.

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Dr.V. Goutham	**Dr. Shaik Abdul Nabi**
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Mr.P. Nagaraj	

SREYAS INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I, **NAGULA SOWMYA** bearing Hall ticket No. **17VE1A05F6** hereby declare that the Mini Project titled "A COMPREHENSIVE STUDY ON SOCIAL NETWORK MENTAL DISORDERS DETECTION VIA ONLINE SOCIAL MEDIA MINING" done by us under the guidance of **Mr. P .NAGARAJ**, which is submitted in the partial fulfillment of the requirement for the award of the B.Tech degree in **Computer Science and Engineering Branch** at **Sreyas Institute of Engineering & Technology** for Jawaharlal Nehru Technological University, Hyderabad is our original work.

NAGULA SOWMYA (17VE1A05F6)

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NAGULA SOWMYA (17VE1A05F6)

ABSTRACT

The explosive growth in popularity of social networking leads to the problematic usage. An increasing number of social network mental disorders (SNMDs), such as Cyber-Relationship Addiction, Information Overload, and Net Compulsion, have been recently noted. Symptoms of these mental disorders are usually observed passively today, resulting in delayed clinical intervention. In this paper, we argue that mining online social behaviour provides an opportunity to actively identify SNMDs at an early stage. It is challenging to detect SNMDs because the mental status cannot be directly observed from online social activity logs. Our approach, new and innovative to the practice of SNMD detection, does not rely on self-revealing of those mental factors via questionnaires in Psychology. Instead, we propose a machine learning framework, namely, Social Network Mental Disorder Detection (SNMDD), that exploits features extracted from social network data to accurately identify potential cases of SNMDs. We also exploit multi-source learning in SNMDD and propose a new SNMD-based Tensor Model (STM) to improve the accuracy. To increase the scalability of STM, we further improve the efficiency with performance guarantee. Our framework is evaluated via a user study with 3126 online social network users. We conduct a feature analysis, and also apply SNMDD on large-scale datasets and analyse the characteristics of the three SNMD types. The results manifest that SNMDD is promising for identifying online social network users with potential SNMDs.

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CHAPTER-1

INTRODUCTION

1.1 PROBLEM STATEMENT

The explosive growth in popularity of social networking leads to the problematic usage. An increasing number of social network mental disorders SNMDs such as Cyber-Relationship Addiction, Information Overload, and Net Compulsion, have been recently noted. Symptoms of these mental disorders are usually observed passively today, resulting in delayed clinical intervention. In this paper, we argue that mining online social behavior provides an opportunity to actively identify SNMDs at an early stage. It is challenging to detect SNMDs because the mental status cannot be directly observed from online social activity logs. Our approach, new and innovative to the practice of SNMD detection, does not rely on self-revealing of those mental factors via questionnaires in Psychology.

1.2 MOTIVATION

The research reveals that social network addiction may negatively impact emotional status, causing higher hostility, depressive mood, and compulsive behavior which leads to actively identify SNMDs at an early stage.

1.3 SCOPE

Routine recording of data on SNMDs takes place at an early stage of this process, providing an opportunity to use machine learning to predict future stages in the process which also helps us to find whether a person is normal or mentally disordered.

1.4 OUTLINE

The outline of this project is to mainly develop a web application that allows the admins to identify the mental status of a person based on the posts and comments of the users. The admins can view the users friend requests and responses as well.

CHAPTER-2

LITERATURE SURVEY

2.1 EXISTING SYSTEM

- Chang et. al [43] employ an NLP-based approach to collect and extract linguistic and content-based features from online social media to identify Borderline Personality Disorder and Bipolar Disorder patients.

- Saha et al. [44] extract the topical and linguistic features from online social media for depression patients to analyze their patterns.

- Choudhury et al. [45] analyze emotion and linguistic styles of social media data for Major Depressive Disorder (MDD). However, most previous research focuses on individual behaviors and their generated textual contents but do not carefully examine the structure of social networks and potential Psychological features.

DISADVANTAGES:

- There are no temporal behavior features to trace Mental Disorders.
- There are no techniques for offline interaction.
- Peak memory usage during detection.
- Due to high computation time, detection approach becomes complex.

2.2 PROPOSED SYSTEM

- In the proposed system, the system aims to explore data mining techniques to detect three types of SNMDs [1]: 1) Cyber-Relationship (CR) Addiction, which includes the addiction to social networking, checking and messaging to the point where social relationships to virtual and online friends become more important than real-life ones with friends and families; 2) Net Compulsion (NC), which includes compulsive online social gaming or gambling, often resulting in financial and job-related problems; and 3) Information Overload (IO), which includes addictive surfing of user status and news feeds, leading to lower work productivity and fewer social interactions with families and friends offline.

- Two key challenges exist in design of SNMDD: i) we are not able to directly extract mental factors like what have been done via questionnaires in Psychology and thus need new features for learning the classification models ii) we aim to exploit user data logs from multiple OSNs and thus need new techniques for integrating multi-source data based on SNMD characteristics.

ADVANTAGES:

- The system develops a machine learning framework to detect SNMDs, called Social Network Mental Disorder Detection (SNMDD).
- Social diversity-based features (SDiv) Researchers have observed that diversity improves the depth of people thinking for both majority or minority

CHAPTER-3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 INRODUCTION

3.1.1 Purpose

Mental health is the most predominant factor that is mandatory for every human being in this world. Without mental health neither achievement nor success comes in one's way. It affects how we think, feel, and act as we cope with life. It also helps determine how we handle stress, relate to others, and make choices. Mental health is important at every stage of life, from childhood and adolescence through adulthood and aging which paves way to detecting SNMDs via online social media mining.

3.1.2 Document Conventions

This Document was created based on the IEEE template for System Requirement Specification Documents.

3.1.3 Intended Audience and Reading Suggestions

The major stakeholders of the project are Admin and the Users. Users can register and login into the system once their registration is authorised by the admin. Once the admin has logged into the system, he/she can upload any posts and also comment on others posts as well. The admin can view users and authorise, add filter category, view all mentally disorders detection and normal users based on posts and comments, view user search transactions etc.

3.1.4 Product Scope

The core function of SOCIAL NETWORK MENTAL DISORDER DETECTION framework is to detect mentally disordered people at a rudimentary stage. It is challenging to identify SNMDs because the mental status cannot be directly observed from online social activity logs. This

approach, new and innovative to the practice of SNMD detection, does not rely on self-revealing of those mental factors via questionnaires in Psychology.

3.1.5 References

INTERNET, INSTRUCTOR

3.2. OVERALL DESCRIPTION

3.2.1 Product Perspective

A COMPREHENSIVE STUDY ON SOCIAL NETWORK MENTAL DISORDERS DETECTION VIA ONLINE SOCIAL MEDIA MINING helps us to distinguish between normal OSN users and mentally disordered OSN users, therefore detecting the abnormal users at an early and reducing the escalating number of such users.

3.2.2 Product Functions

The application has main menu, from which the rest of the features or functions can be accessed. These main features as listed are:

1. Admins area

2. Users area

3.2.3 User Classes and Characteristics

Our application hopes to draw on two main groups. The Admins and Users have their own individual login id and password. The two main user groups are:

User: who can upload posts and comments depending on his mood and can also check his own profile and the profile of his friends as well.

Admin: who can view the list of users and authorize them and can further add filter category and perform many more functions depending upon the requirement.

3.2.4 Operating Environment

SOFTWARE REQUIREMENTS

- Operating System : Windows 7, Windows 8,Windows 10(x32,x64)
- Front End : HTML , CSS
- Back End : Java Script
- Database : My SQL

HARDWARE REQUIREMENTS

- Processor : i5 processor
- Speed : 2.9 GHz (Minimum).
- RAM : 4 GB (Minimum).
- Hard Disk : 4 GB (Minimum).

3.2.5 Design and Implementation Constraints

The Programming language shall be MYSQL for the database.

The users working system should have an internet.

3.2.6 User Documentation

The product is under development stage and requires a complete implemented prototype to explain the user documentation. Once the prototype is designed and implemented, online manuals, user manuals can be provided.

3.2.7 Assumptions and Dependencies

It is assumed that the permission will be obtained from different departments to link the system with SNMDD.

3.3 EXTERNAL INTERFACE REQUIREMENTS

3.3.1 User Interface

The interface shall be web based allowing user to access the system. This framework will be very clear and interactive to the user. There will be a proper collection of GUI interface, which will provide better look and be user friendly.

3.3.2 Hardware Interface

The hardware requirement at the user end is really simple and the website can also run on the hardware that can run a basic simple browser. The application will run on any system with basic requirement.

3.3.3 Software Interface

During our system development, we design both static and dynamic website interfaces. The application software will run on operating system Window 8, windows 10(x32, x64) with additional requirements.

3.3.4 Communication Interface

The application supports Google Chrome, Mozilla Firefox web browsers.

3.4 FUNCTIONAL REQUIREMENTS

User:

In this module, the user can upload posts and comments depending on his mood and can also check his own profile and the profile of his friends as well.

Admin:

In this module, the admin can view the list of users and authorize them and can further add filter category and perform many more functions depending upon the requirement.

3.5 NON-FUNCTIONAL REQUIREMENTS

3.5.1 Performance Requirements

The performance requirements refer to static numerical requirements placed on the interaction between the users and the software.

Response Time:

Average response time shall be less than 5 seconds

Recovery Time:

In case of system failure, redundant system shall resume operations within 30 seconds.
Average repair time shall be less than 1 hr.

Start-Up/Shutdown Time:

The system shall be operational within 2 minutes of starting up.

Capacity:

The system accommodates large sets of data.

Utilization of Resources:

The system shall store in the database no more than two million queries. If the database grows over this limit, old transaction shall be backed up and deleted from the operational database.

3.5.2 System Resource Consumption

Resource consumption of this application should not reach any amount that renders the website unusable. The application should be capable of operating in the background which should enable the user to utilize other applications.

3.5.3 Safety Requirements

The database may get crashed at any time due to virus or OS failure. Therefore, it is required to take database backup so that database is not lost. The group-based policy for access control of data project will not affect the data stored outside of its server will not affect any other application installed.

3.5.4 Security Requirements

- System will be secured in database.
- System will have different types of users and every user has his/her user id and password.

3.5.5 Software Quality Attributes

Reliability

The system shall be reliable i.e., in case the server crashes, a backup server will be there to work which will be maintained continuously.

Availability

The system will not be available to unauthorised users round the clock i.e., they cannot access the system at any time they want.

Security

The software should never disclose any personal information of any user and should collect no personal information from its own users.

Maintainability

Using an object-oriented approach and proper database modeling along with extensive documentation will make it easy to troubleshoot and maintain in the future.

Usability

The interfaces of the system will be user friendly enough that every user will be able to use it easily.

CHAPTER-4

ARCHITECTURE

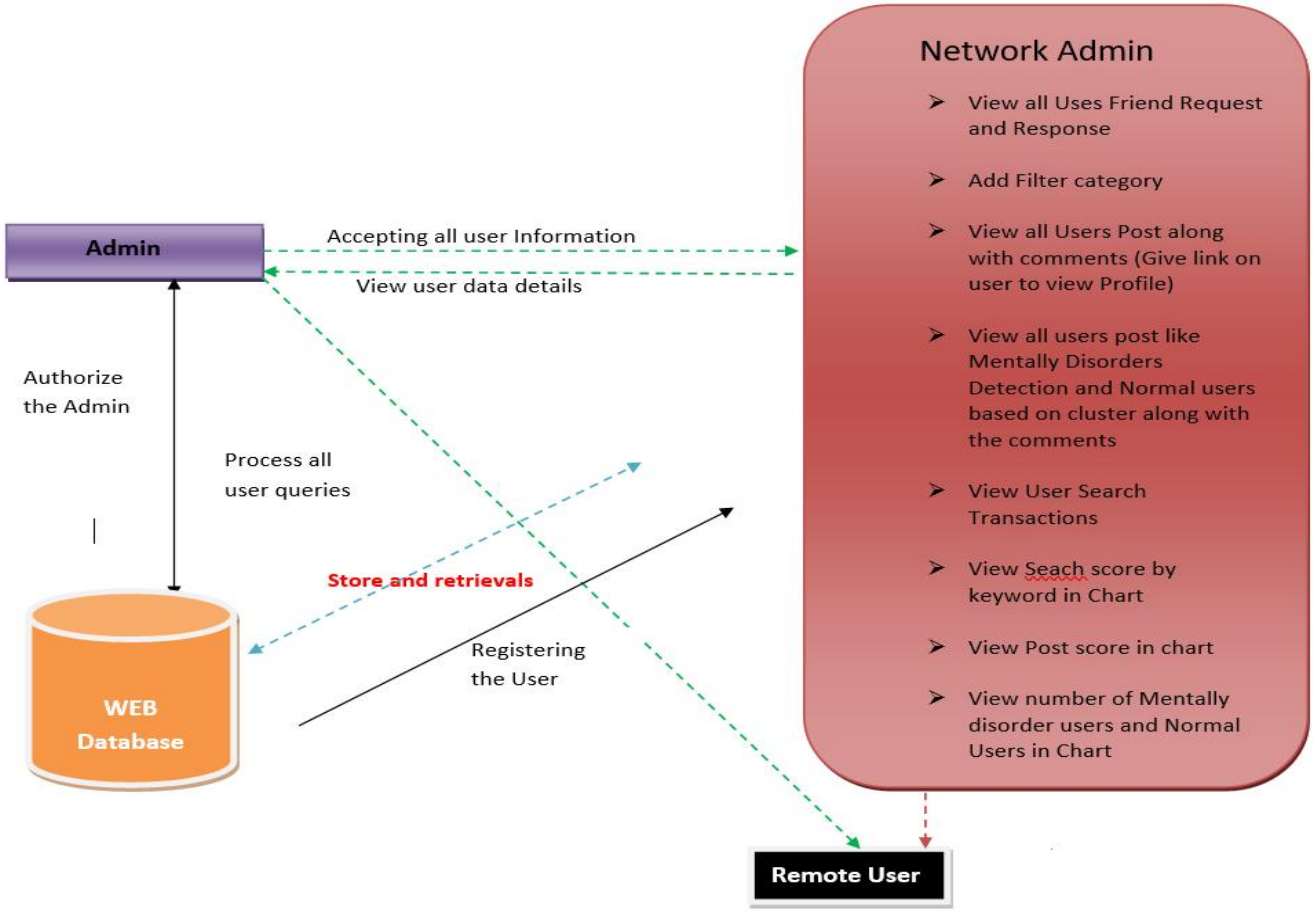


Fig 4.1 Architecture Diagram

User:

In this module, the user can upload posts and comments depending on his mood and can also check his own profile and the profile of his friends as well.

Admin:

In this module, the admin can view the list of users and authorize them and can further add filter category and perform many more functions depending upon the requirement.

CHAPTER-5

SYSTEM DESIGN

5.1 IMPORTANCE OF DESIGN

The purpose of the design phase is to plan a solution of the problem specified by the requirement document. It is the process of defining software methods, functions, objects and overall structure and interaction of your code so that the resulting functionality will satisfy your users requirements. It allows you to do the best abstraction, to understand the requirements better and meet them better. This prevents redundancy and increases reusability.

This phase is the first step in moving from the problem domain to the solution domain. In other words, starting with what is needed design takes us towards how to satisfy the needs. The design of a system is perhaps the most critical factor affection the quality of the software; it has a major impact on the later phase, particularly testing, maintenance. The output of this phase is the design document. This document is similar to a blueprint for the solution and is used later during implementation, testing and maintenance. The design activity is often divided into two separate phases System Design and Detailed Design.

System Design also called top-level design sign aims to identify the modules that should be in the system, the specifications of these modules, and how they interact with each other to produce the desired results. During, detailed design, the internal logic of each of the modules specification in system design is decided. During this phase, the details of the data are usually specified in a high-level design description language, which is independent of the target language in which the software will eventually be implemented.

In system design the focus is on identifying the modules, whereas during detailed design the focus is on designing the logic for each of the modules. During the system design activities, Developers bridge the gap between the requirements specification, produced during requirements elicitation and analysis, and the system that is delivered to the user.

5.2 UML DIAGRAMS

- The unified modeling language allows the software engineer to express an analysis model using the modeling notation that is governed by a set of syntactic, semantic and pragmatic rules
- A UML system is represented using five different views that describe the system from distinctly different perspective. Each view is defined by a set of diagrams, which is as follows.

User Model View

- This view represents the system from the user's perspective.
- The analysis representation describes a usage scenario from the end-user's perspective.

Structural Model View

- In this model the data and functionality are arrived from inside the system.
- This model view models the static structures.

Behavioral Model View

- It represents the dynamic behavior of parts of the system, depicting the interactions of collection between various structural elements described in the user model and structural model view.

Implementation Model View

- In this the structural and behavioral parts of the system are represented as they are to be built.

Environmental Model View

- In this the structural and behavioral aspects of the environment in which he system is to be implemented and represented.
- UML is specifically constructed through two different domains. They are
  - ❖ UML Analysis modeling, this focuses on the user model and structural model views of the system.
  - ❖ UML design modeling, which focuses on the behavioral modeling, implementation modeling and environmental model views.
- Use case diagrams represent the functionality of the system from a user's point of view. Use cases are used during requirements elicitation and analysis to represent the functionality of the system. Use cases focus on the behavior of the system from external point of view.
- Actors are external entities that interact with the system. Examples of actors include users like administrator, bank customer…etc…, or another system like central database.

5.2.1 USE CASE DIAGRAMS:

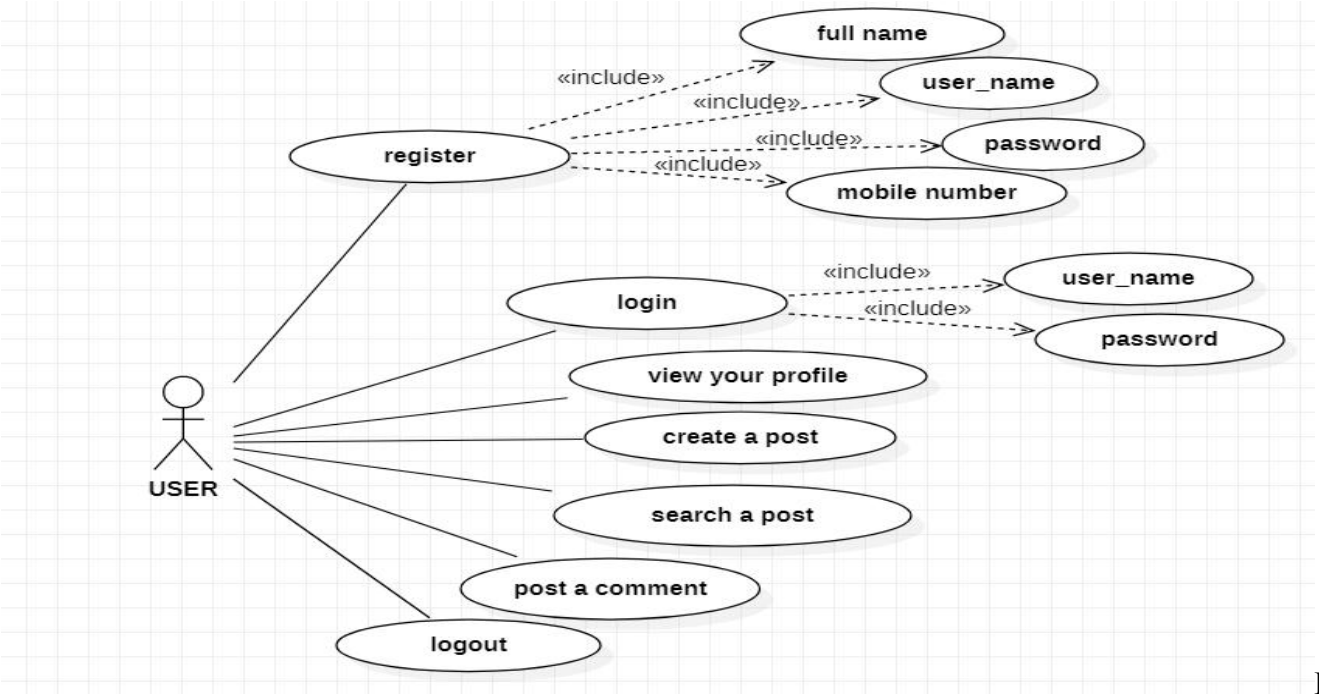


Fig 5.1 Use case Diagram for User

Description: The user has to first register and then login into the application using his/her login credentials. He can then view his/her profile, create a post, search for a post, post a comment, perform many other functions and then finally logout.

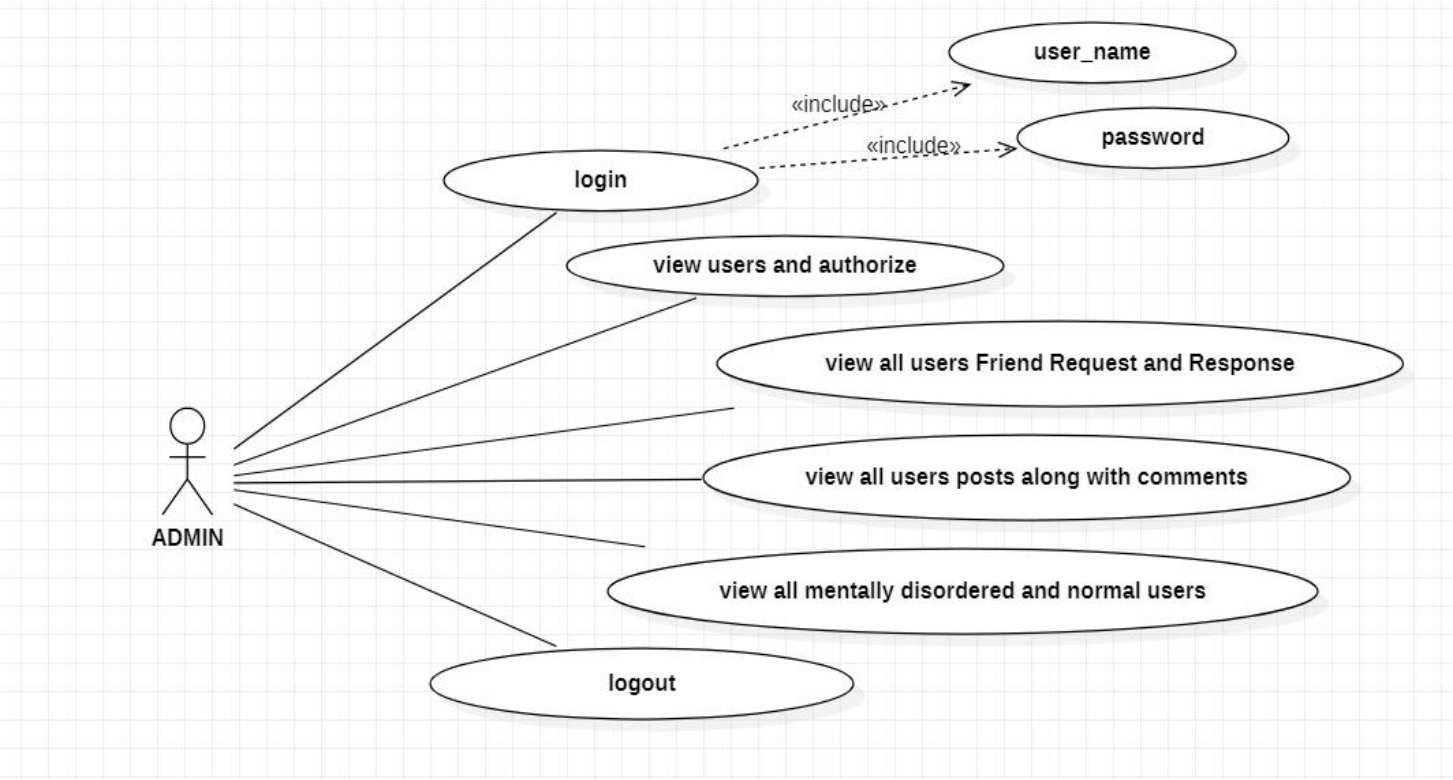


Fig 5.2 Use case diagram for Admin

Description: The admin has to login using his username and password after which he can view users and authorize, view all users friend requests and responses, view users posts along with comments, view all mentally disordered and normal users, etc and then logout.

5.2.2 CLASS DIAGRAM

Class diagram describe the structure of the system in terms of classes and objects and their relationships.

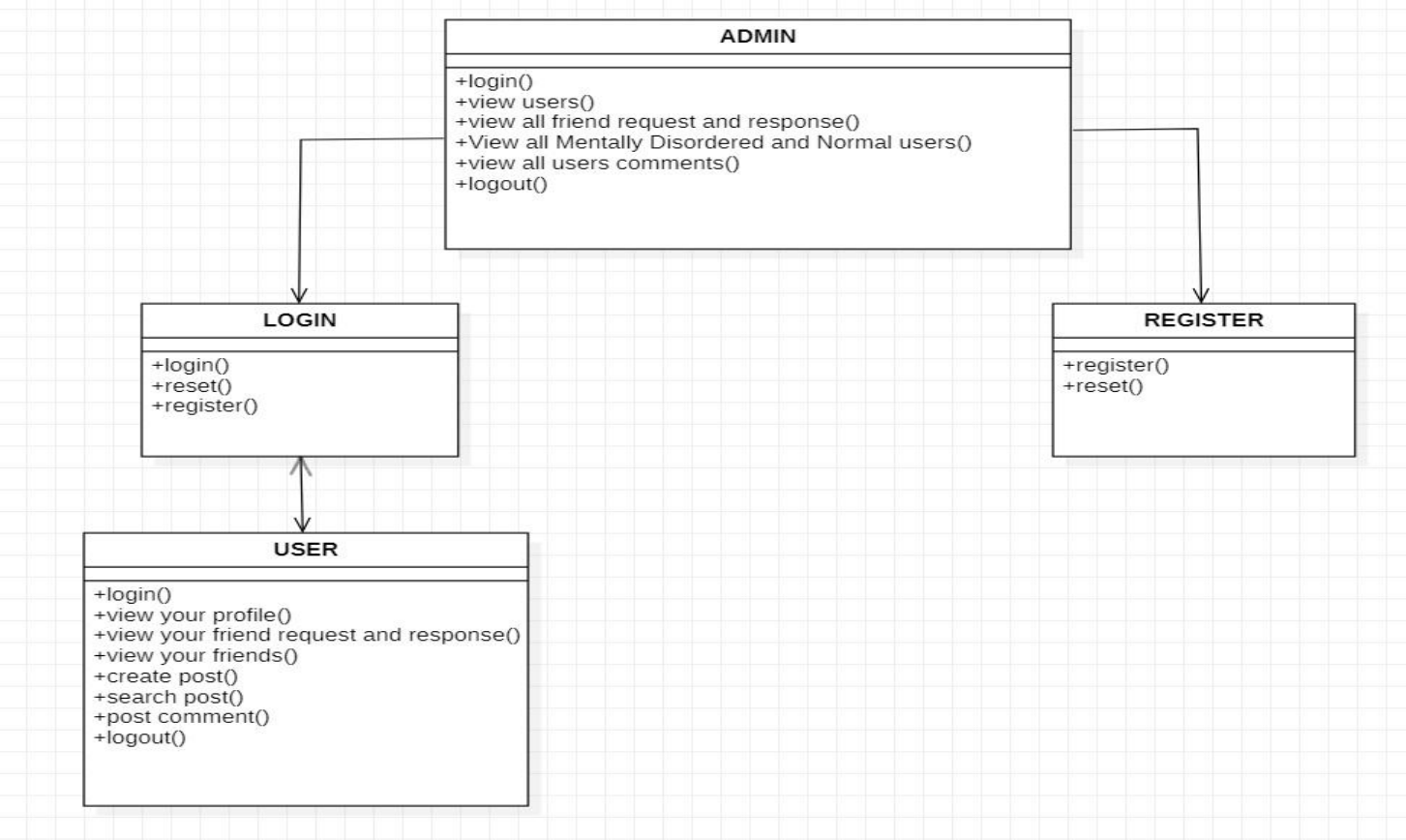


Fig 5.3 Class diagram

Classes:

- Login
- Register
- Admin
- User

5.2.3 SEQUENCE DIAGRAM

A sequence graphically depicts how objects interact with each other via messages in the execution of a use case or operation. They illustrate how messages are sent and received between objects and the sequence of message transfer. It also details how operations are carried out according to the time of operation.

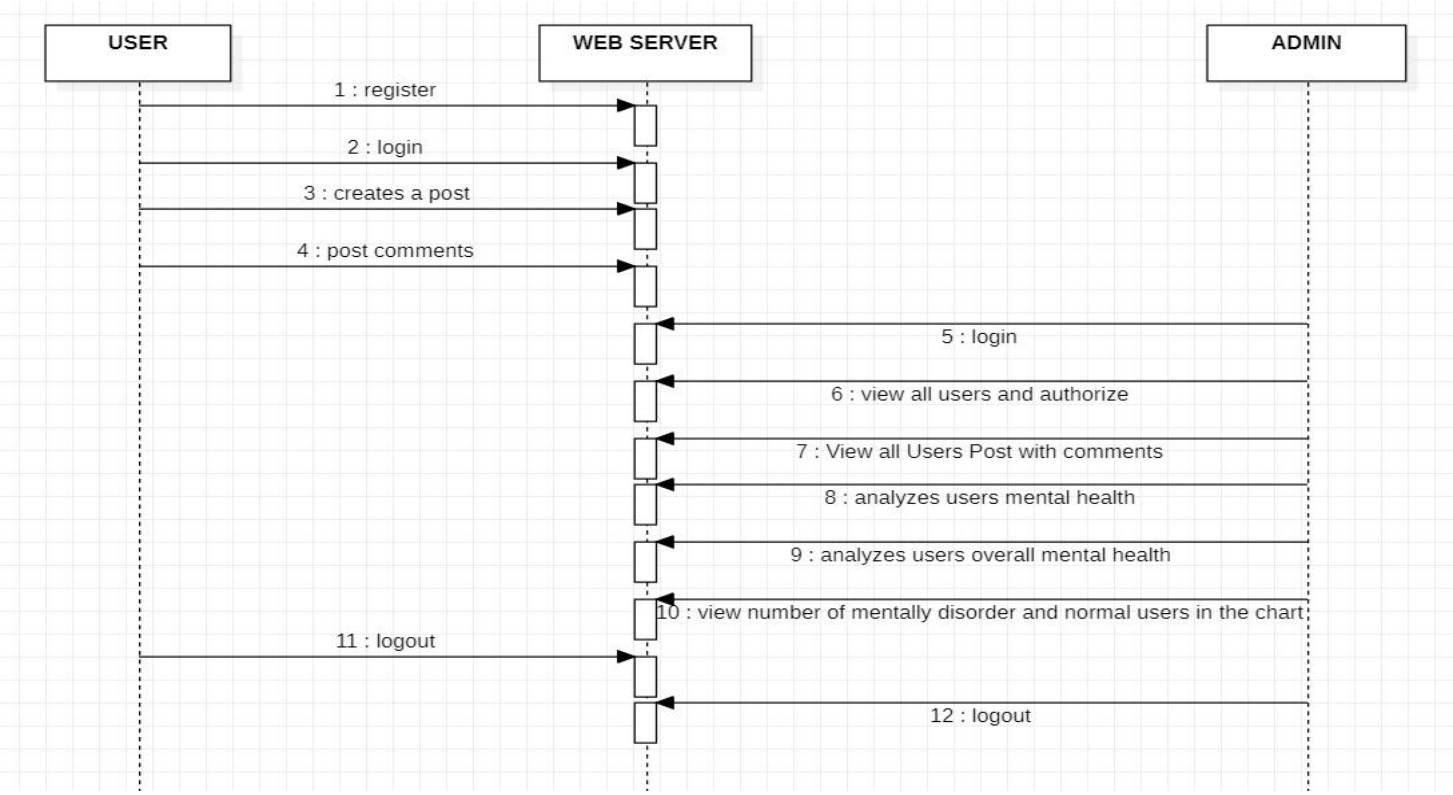
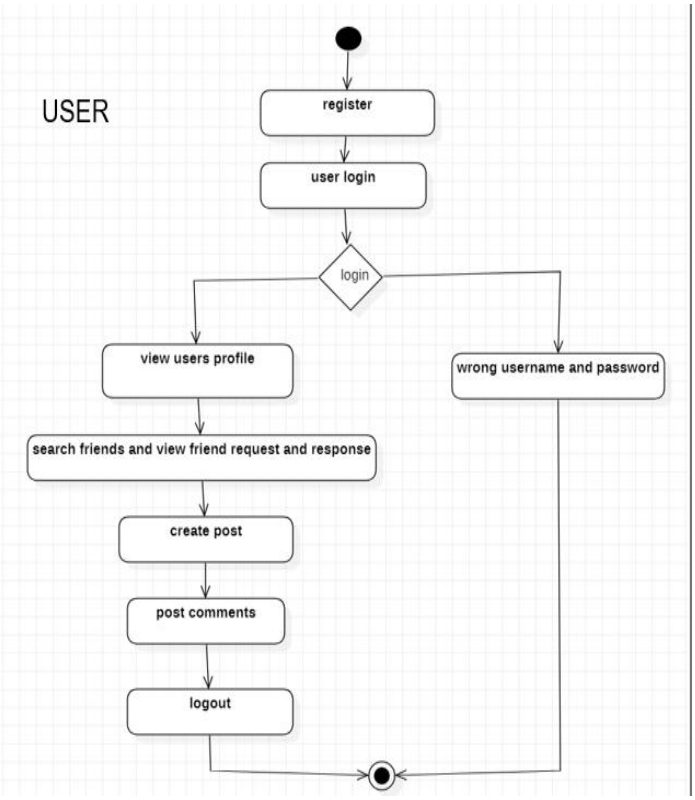


Fig 5.4 Sequence diagram

5.2.4 ACTIVITY DIAGRAM

Activity diagrams graphically depict the sequential flow of activities of either a business processor a use case. They can also be used to model actions that will be performed when an operation is executed as well as the results of those actions. They focus on the flow of activities involved in a single process. The activity diagram shows how those activities depend on one another.



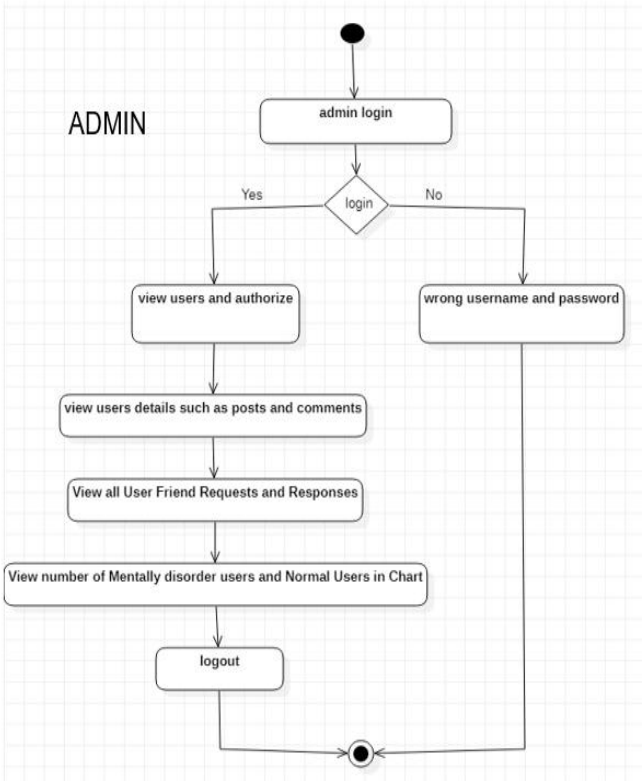


Fig 5.5 Activity diagram

CHAPTER-6

IMPLEMENTATION

6.1 MODULE DESCRIPTION

Implementation includes all those activities that take place to covert from old system to new system. The old system consists of manual operations, which is operated in a very difficult manner from the proposed system. A proper implementation is essential to provide a reliable system to meet the requirements of the organization.

The modules of the application are as follow:

USER: The user has to register first and then has to login unlike the admin. He/she can upload posts and comments, send and accept friend requests and responses and then logout.

ADMIN: The admin can directly login into his account and view users and authorize them. He admin can also view the new registrations of the users. He can also view search score by keyword in chart and also view post score in chart.

6.2 SAMPLE CODE

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Home Page</title>
<meta http-equiv="content-type" content="text/html; charset=utf-8" />
<link href="style.css" rel="stylesheet" type="text/css" />
<script type="text/javascript" src="js/cufon-yui.js"></script>
<script type="text/javascript" src="js/arial.js"></script>
<script type="text/javascript" src="js/cuf_run.js"></script>
<style type="text/css">
<!--
.style2 {font-size: 18px}
.style3 {
    font-size: 24px;
    color: #FF0000;
    font-weight: bold;
}
.style4 {color: #FF0000}
.style5 {color: #FF0000; font-weight: bold; }
-->
</style>
</head>
<body>
<div class="main">
  <div class="header">
    <div class="header_resize">
      <div class="logo">
        <h1><a href="index.html"><span class="style2">A Comprehensive Study on Social Network Mental Disorders Detection via Online Social Media Mining</span></a></h1>
      </div>
      <div class="menu">
        <ul>
          <li><a href="index.html" class="active"><span>Home</span></a></li>
          <li><a href="SocialLogin.jsp"><span>Social Network</span></a></li>
          <li><a href="UserLogin.jsp"><span>User</span></a></li>
          <li><a href="contact.html"><span></span></a></li>
        </ul>
      </div>
      <div class="clr"></div>
    </div>
    <div class="headert_text_resize"> <img src="images/text_area_img.jpg" alt="" width="395" height="396" />
      <div class="textarea">
        <h2>Tensor factorization acceleration, online social network, mental disorder detection, feature extraction</h2>
      </div>
      <div class="clr"></div>
    </div>
    <div class="clr"></div>
  </div>
  <div class="body">
    <div class="body_resize">
      <div class="left">
        <h2>A Comprehensive Study on Social Network<br />
          Mental Disorders Detection<br />
          <span> Tensor factorization acceleration, online social network, mental disorder detection, feature extraction</span></h2>
        <p><img src="images/Home.png" width="446" height="176" /></p>
```

```
55          <p align="justify" class="style5">
</p>
56          <h2 align="justify" class="style4"> </h2>
57        </div>
58        <div class="right">
59          <h2>Sidebar Menu</h2>
60          <ul>
61            <li><a href="#">Home</a></li>
62            <li><a href="SocialLogin.jsp">Social Network </a></li>
63            <li><a href="UserLogin.jsp">User Login </a></li>
64          </ul>
65
66        </div>
67        <div class="clr"></div>
68      </div>
69    </div>
70    <div class="FBG"></div>
71    <div class="footer">
72      <div class="footer_resize">
73        <div class="clr">
74          <div align="center" class="style3">A Comprehensive Study on Social Network Mental Disorders Detection via Online Social Media Mining</div>
75        </div>
76      </div>
77      <div class="clr"></div>
78    </div>
79  </div>
80  <div align=center></div>
81  </body>
82  </html>
```

CHAPTER-7
TESTING

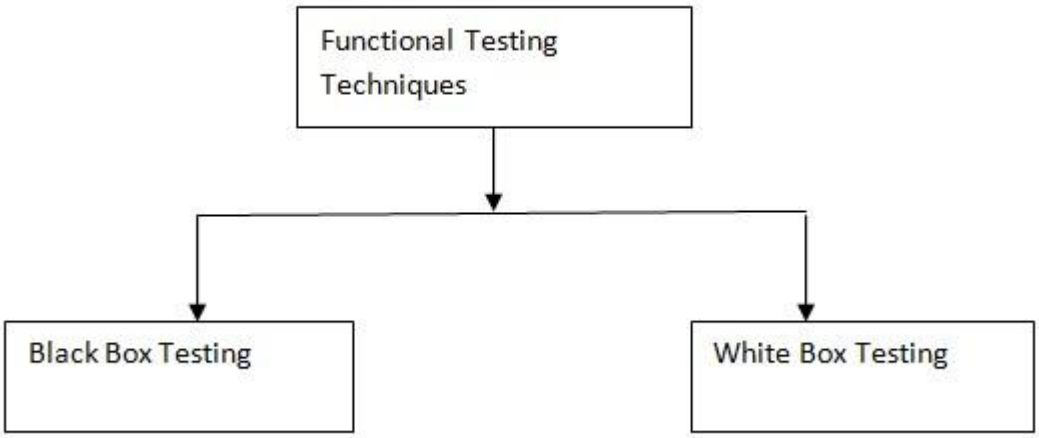
7.1 IMPORTANCE OF TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assembles, assembles and/or a finished product. It is the process of exercising software with the intent of ensuring that the software system its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

7.2 TYPES OF TESTING

7.2.1 FUNCTIONALITY TESTING

Functionality Testing is a testing technique that is used to test the features/functionality of the system or software, should cover all the scenarios including failure paths and boundary cases.
There are two major Functional Testing techniques as shown below:



Test Results:

- All the outgoing links from all the pages to specific domain are valid and working.
- All internal links are working.
- Links redirecting to same web pages are fully functional.
- Data integrity of databases is preserved without redundancies.
- Test links used to send email to admin or other users from web pages are working.

All test cases are successful, no broken links observed.

7.2.2 USABILITY TESTING

Usability testing is a non-functional testing technique that is a measure of how easily the system can be used by end users. It is difficult to evaluate and measure but can be evaluated based on the below parameters:

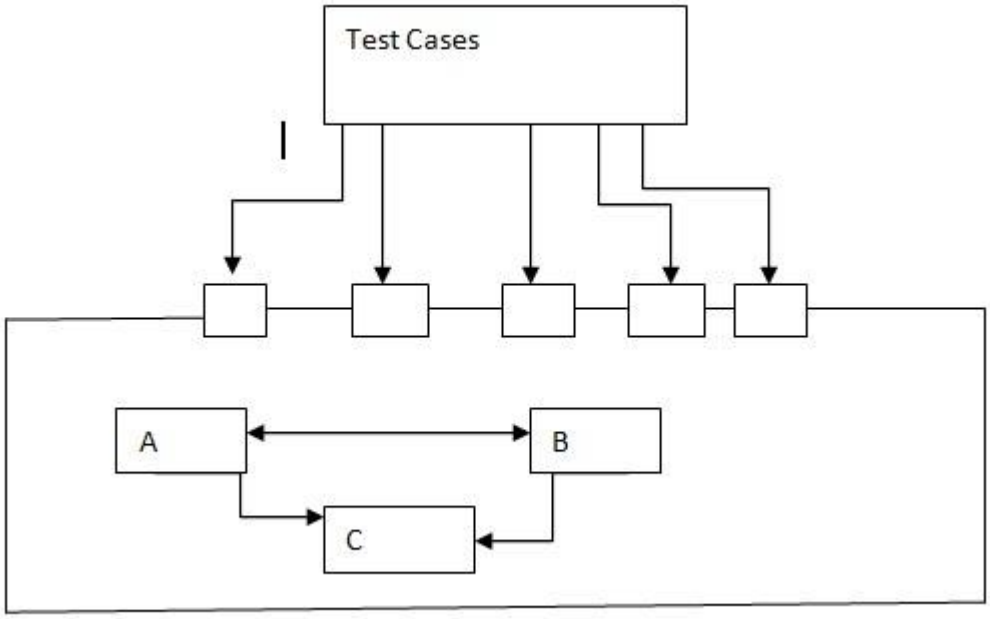
- Level of skill to learn/use the software.
- Time required to get used to in using the software.
- The measure of increase in user productivity if any.
- Assessment of a user's attitude towards using the software.

Test Results:

- The website is highly responsive and intuitive.
- All instructions are provided precisely, satisfying every necessary purpose.
- Main menu is provided on 90% of the web pages.
- Observed high level of consistency.

7.2.3 INTERFACE TESTING

Interface Testing is performed to evaluate whether systems or components pass data and control correctly to one another. It is to verify if all the interactions between these modules are working properly and errors are handled properly.



Test Results:

All the interactions between the web server and the application interface are executed and errors are handled properly. Error messages are displayed appropriately. Executions or connections halted midway are handled properly.

7.2.4 COMPATIBILITY TESTING

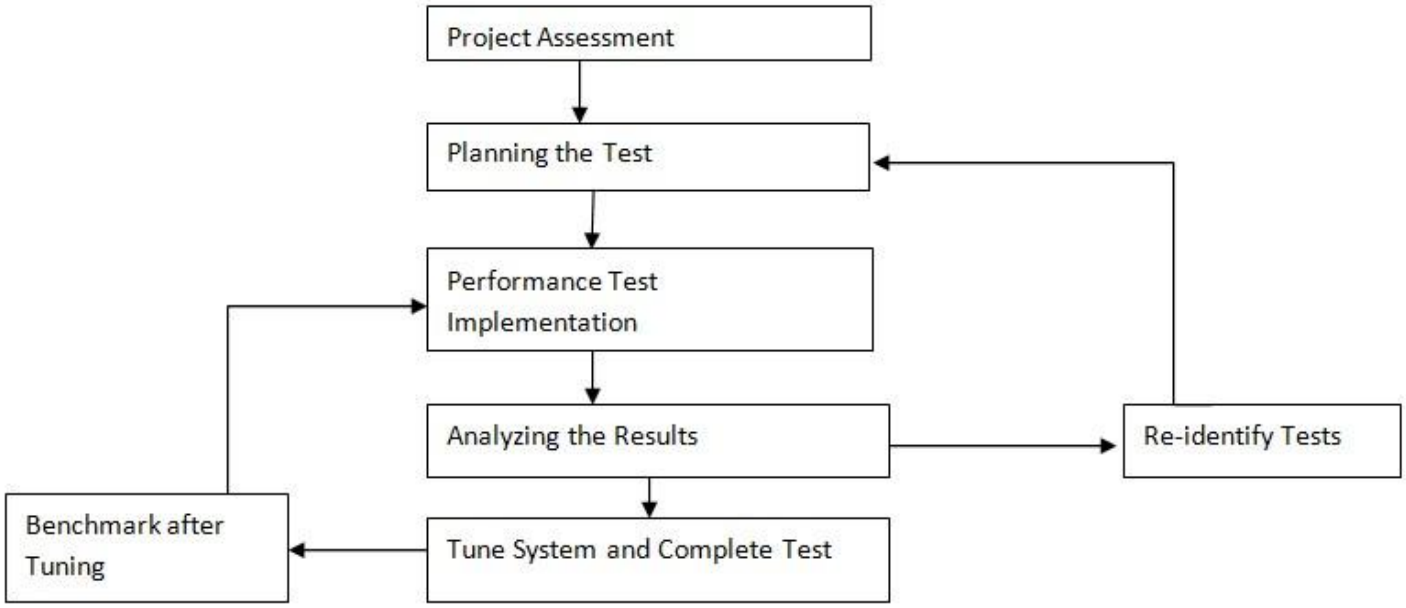
Compatibility testing is a non-functional testing conducted on the application to evaluate the application's compatibility within different environments. It can be of two types – forward compatibility testing and backward compatibility testing.

Test Results:

- Application successfully tested on chrome, Microsoft Edge, Firefox and Safari browsers.
- Application successfully tested on Windows.
- All fonts, page alignment and graphics are getting printed properly.

7.2.5 PERFORMANCE TESTING

Performance testing is a non-functional testing technique performed to determine the system parameters in terms of responsiveness and stability under various workloads. It measures the quality attributes of the system, such as scalability and reliability.



7.3 TEST CASES

+VE TEST CASES:

S .No	Test case Description	Actual value	Expected value	Result
1	Create new user registration process	Enter the personal info and address info.	Update personal info and address info in to oracle database successfully	True
2	Enter the username and password	Verification of login details.	Login Successfully	True
3	Upload file into single cloud , multiple cloud	Enter all fields	Web data uploaded successfully	True
4	Enter keyword query submission	Enter valid query	Display relevant records based keyword query	True

-VE TEST CASES:

S .No	Test case Description	Actual value	Expected value	Result
1	Create the new user registration process	Enter the personal info and address info.	Personal info and address info its not update into database successfully.	False
2	Enter the username and password	Verification of login details.	Login failed	False
3	Upload information	Enter all fields	Web data is not create successfully.	False
4	Enter keyword query submission	Enter valid query	Relevant records are not present in database	False

CHAPTER-8

RESULTS

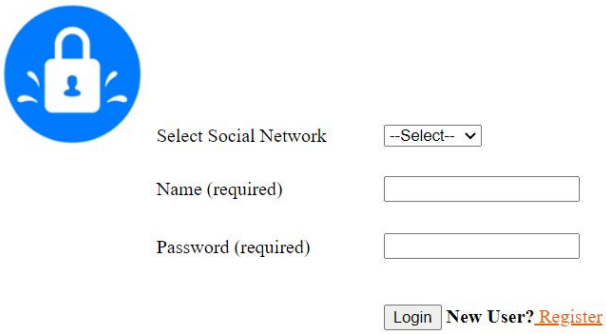


Fig. Admin login

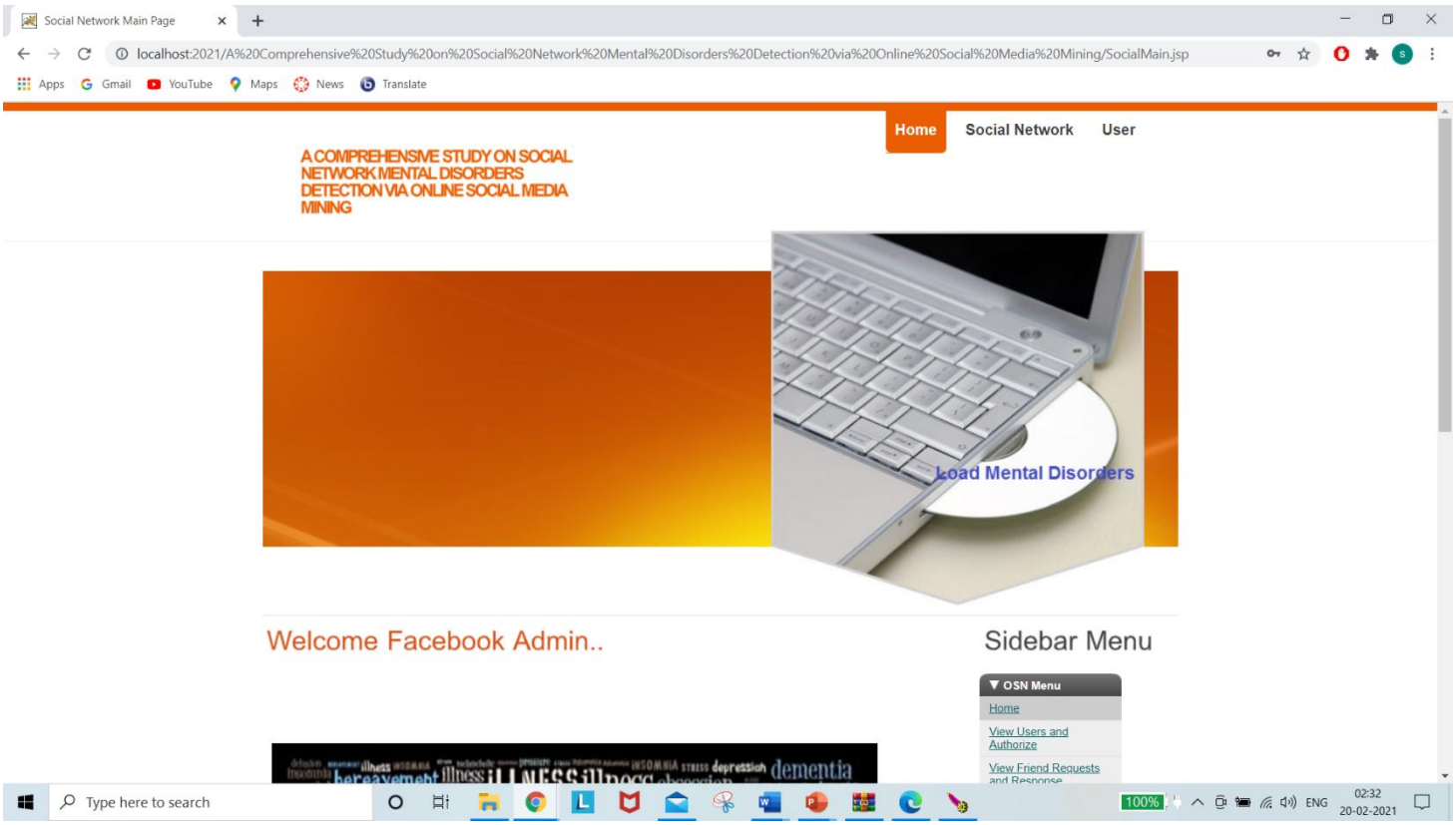


Fig. Admin page

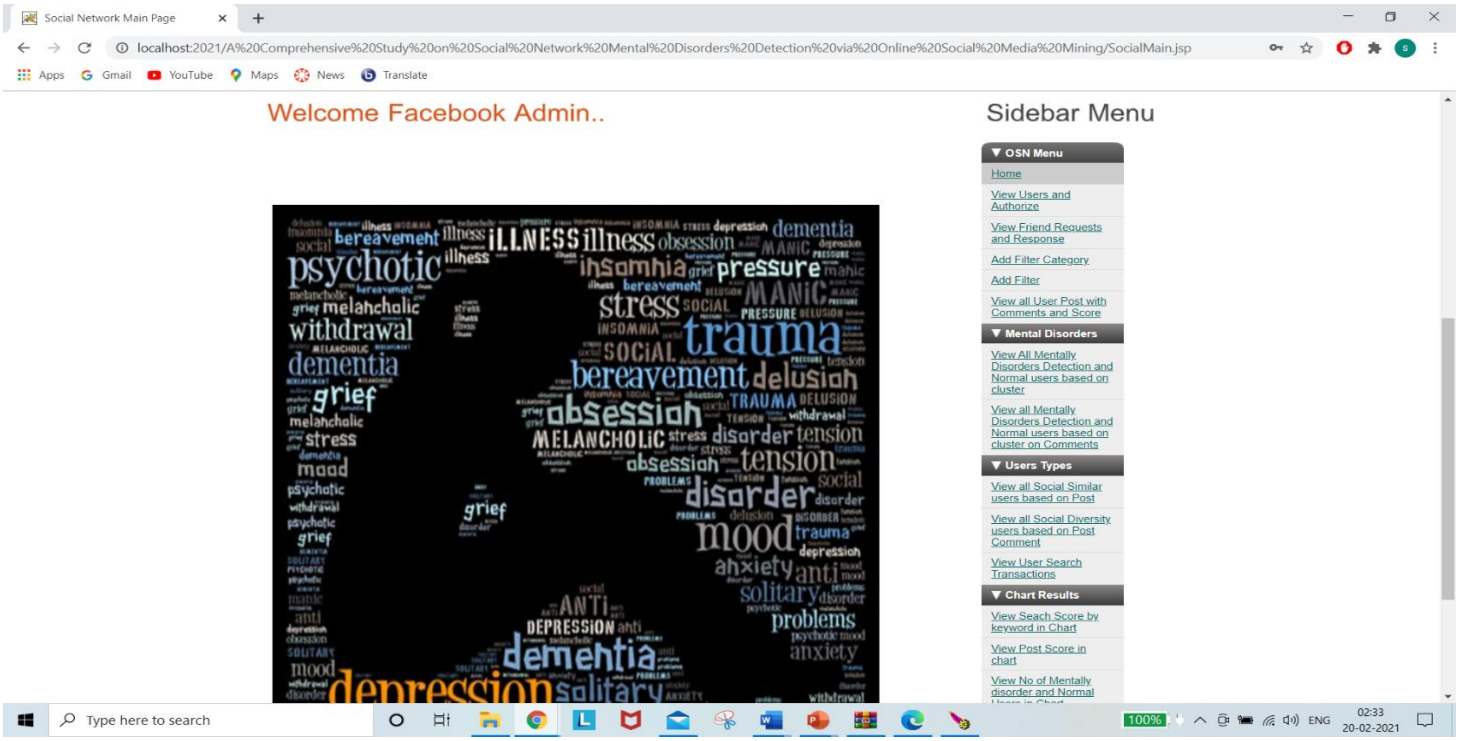
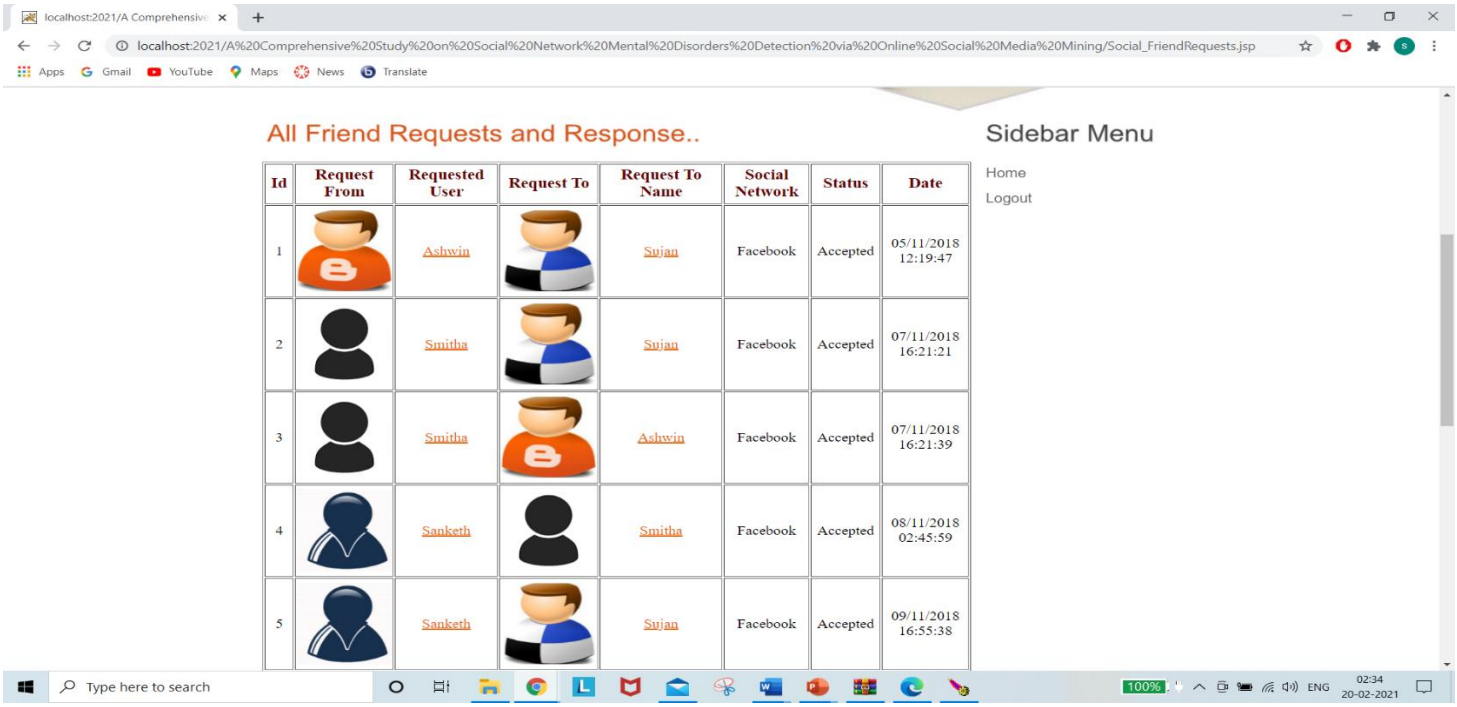


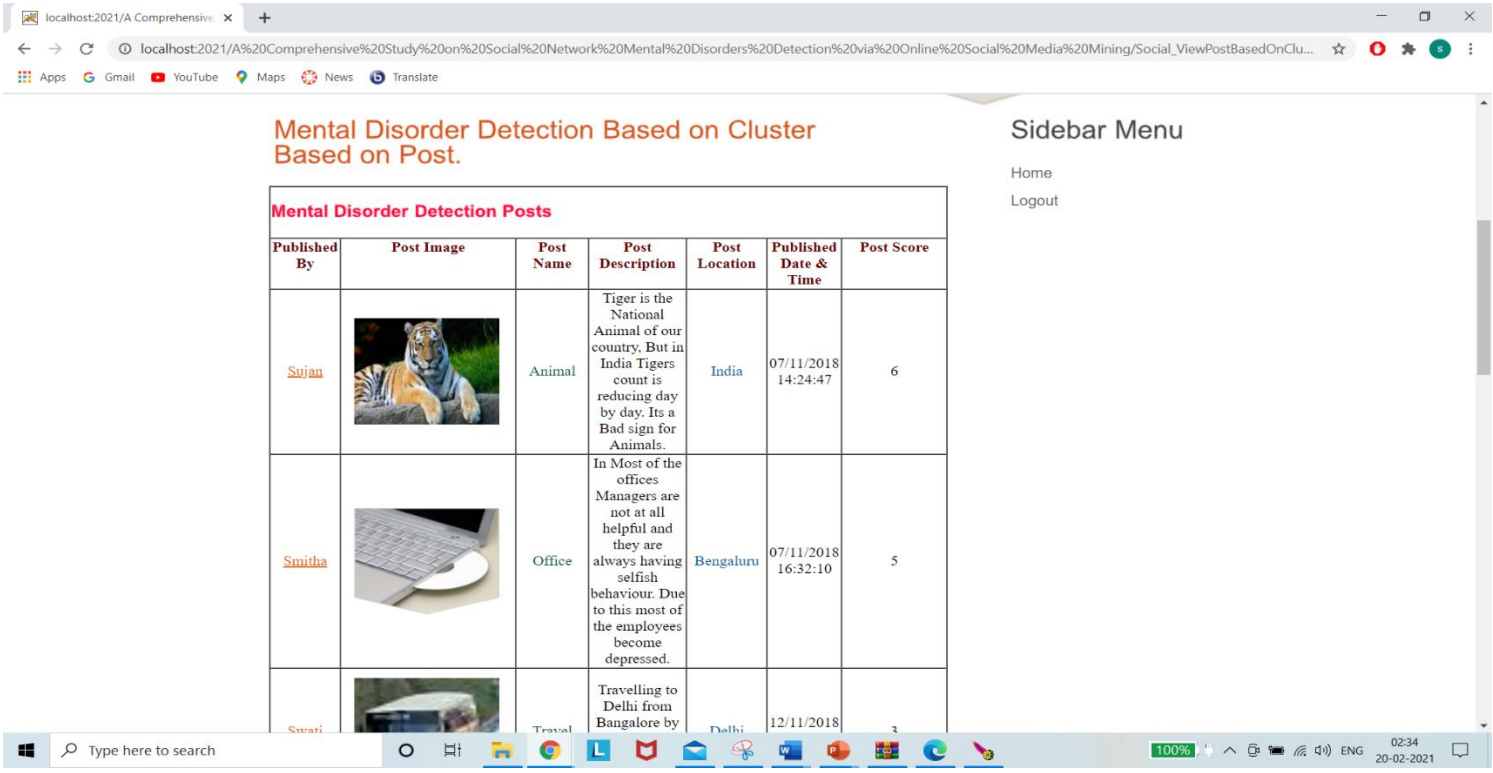
Fig. Functions admin can perform

Fig. Admin can authorize users



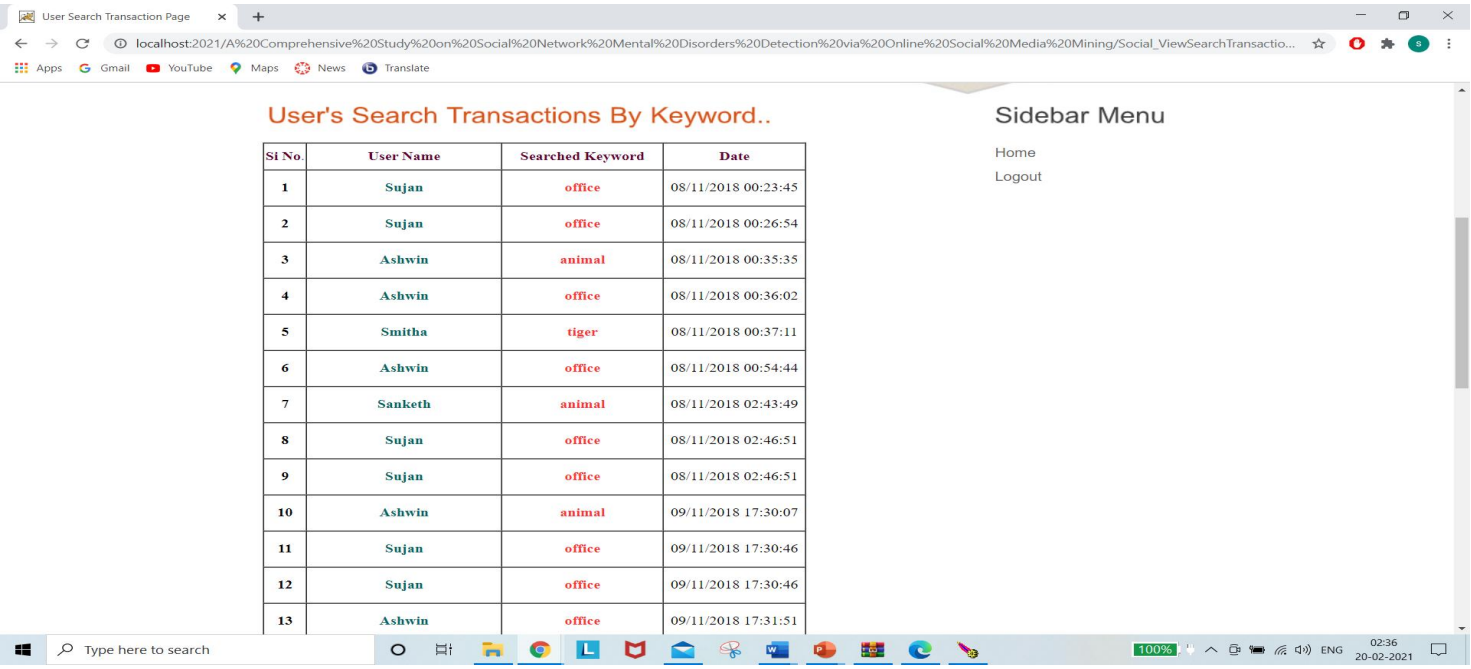
Id	Request From	Requested User	Request To	Request To Name	Social Network	Status	Date
1		Ashwin		Sujan	Facebook	Accepted	05/11/2018 12:19:47
2		Smitha		Sujan	Facebook	Accepted	07/11/2018 16:21:21
3		Smitha		Ashwin	Facebook	Accepted	07/11/2018 16:21:39
4		Sanketh		Smitha	Facebook	Accepted	08/11/2018 02:45:59
5		Sanketh		Sujan	Facebook	Accepted	09/11/2018 16:55:38

Fig. Admin can view friend requests and responses



Published By	Post Image	Post Name	Post Description	Post Location	Published Date & Time	Post Score
Sujan		Animal	Tiger is the National Animal of our country, But in India Tigers count is reducing day by day. Its a Bad sign for Animals.	India	07/11/2018 14:24:47	6
Smitha		Office	In Most of the offices Managers are not at all helpful and they are always having selfish behaviour. Due to this most of the employees become depressed.	Bengaluru	07/11/2018 16:32:10	5
Swati		Travel	Travelling to Delhi from Bangalore by	Delhi	12/11/2018	3

Fig. Admin can view mentally disordered people based on the posts



Si No.	User Name	Searched Keyword	Date
1	Sujan	office	08/11/2018 00:23:45
2	Sujan	office	08/11/2018 00:26:54
3	Ashwin	animal	08/11/2018 00:35:35
4	Ashwin	office	08/11/2018 00:36:02
5	Smitha	tiger	08/11/2018 00:37:11
6	Ashwin	office	08/11/2018 00:54:44
7	Sanketh	animal	08/11/2018 02:43:49
8	Sujan	office	08/11/2018 02:46:51
9	Sujan	office	08/11/2018 02:46:51
10	Ashwin	animal	09/11/2018 17:30:07
11	Sujan	office	09/11/2018 17:30:46
12	Sujan	office	09/11/2018 17:30:46
13	Ashwin	office	09/11/2018 17:31:51

Fig. Admin can view user search transactions

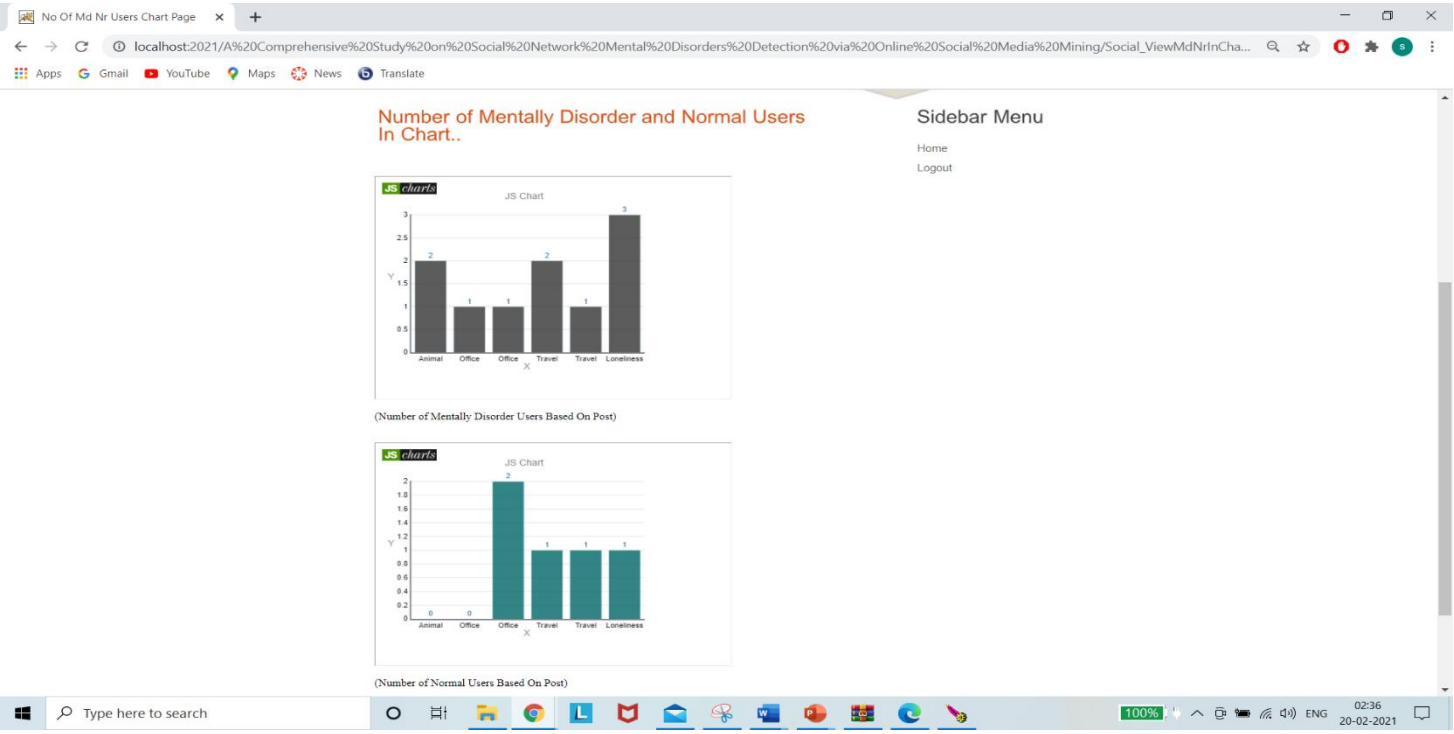


Fig. Admin can view number of mentally disordered and normal users in chart and later logout

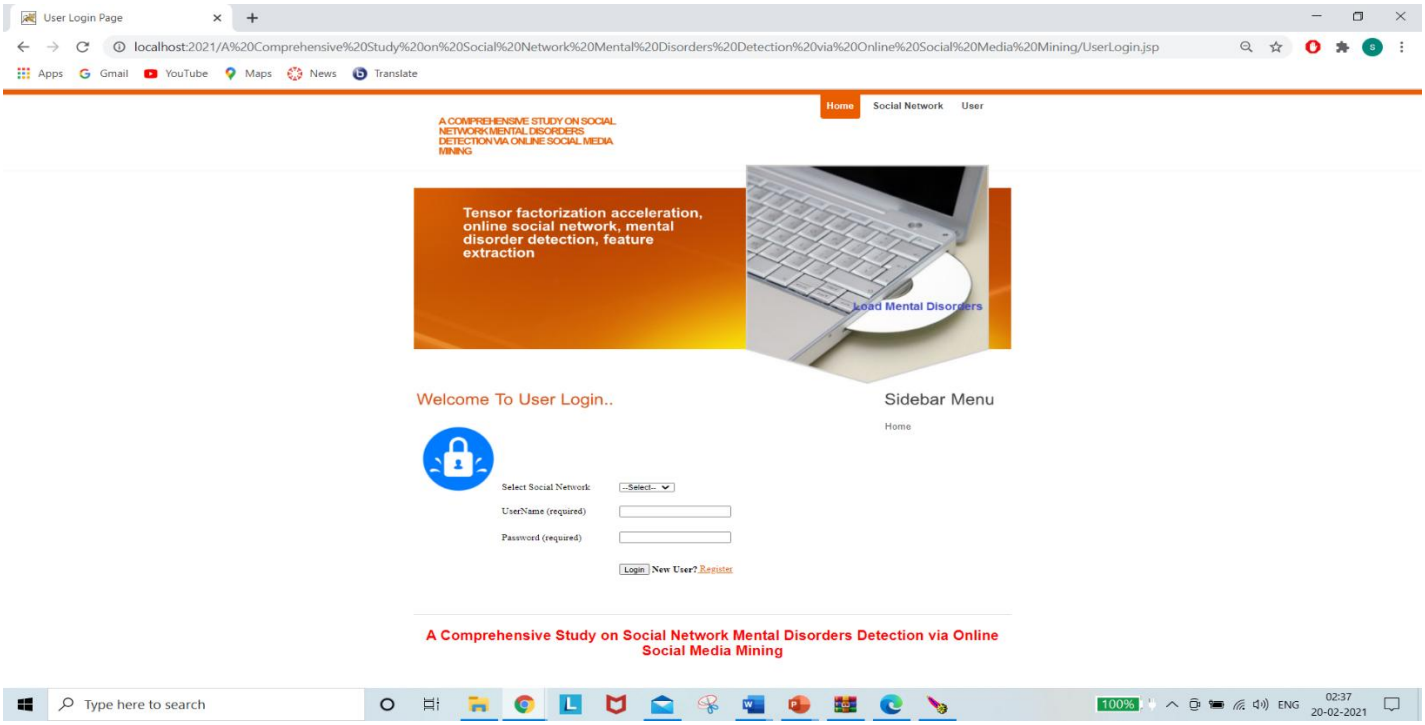


Fig. User login

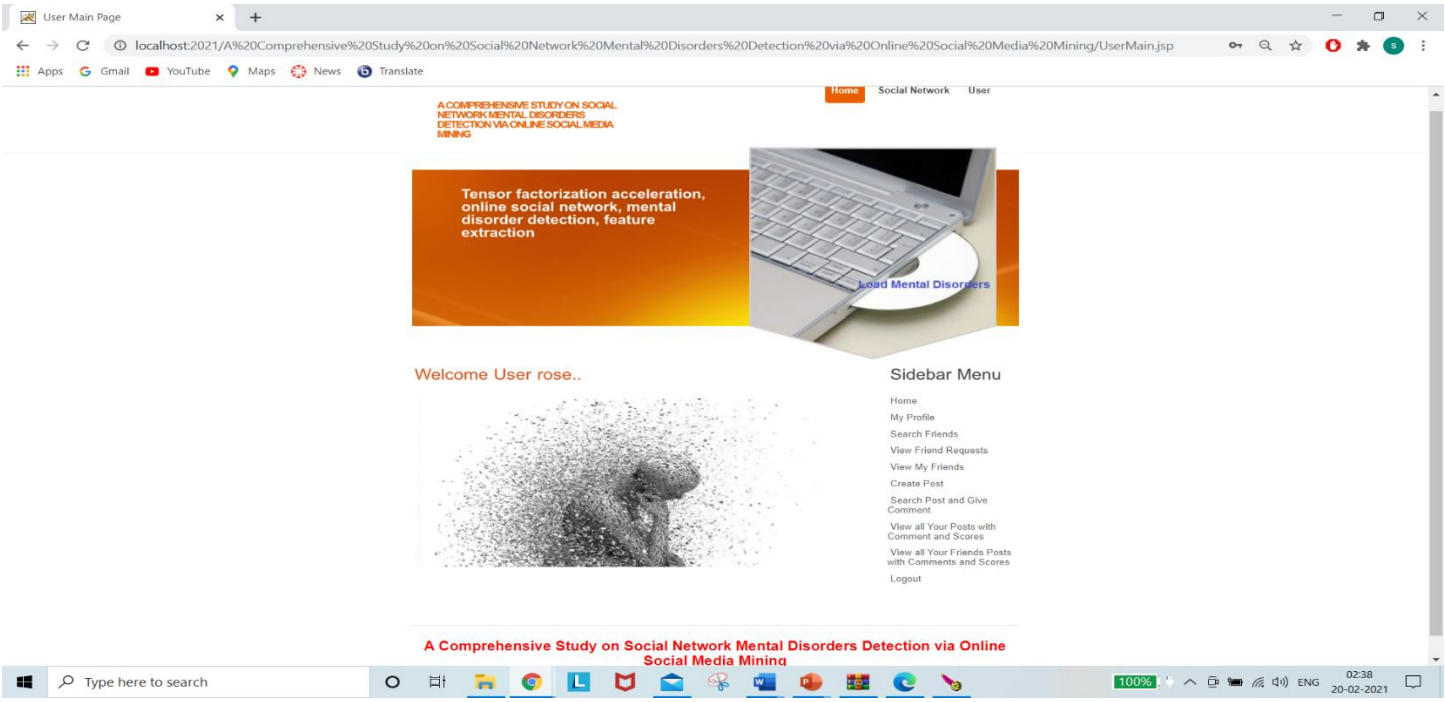


Fig. User home page

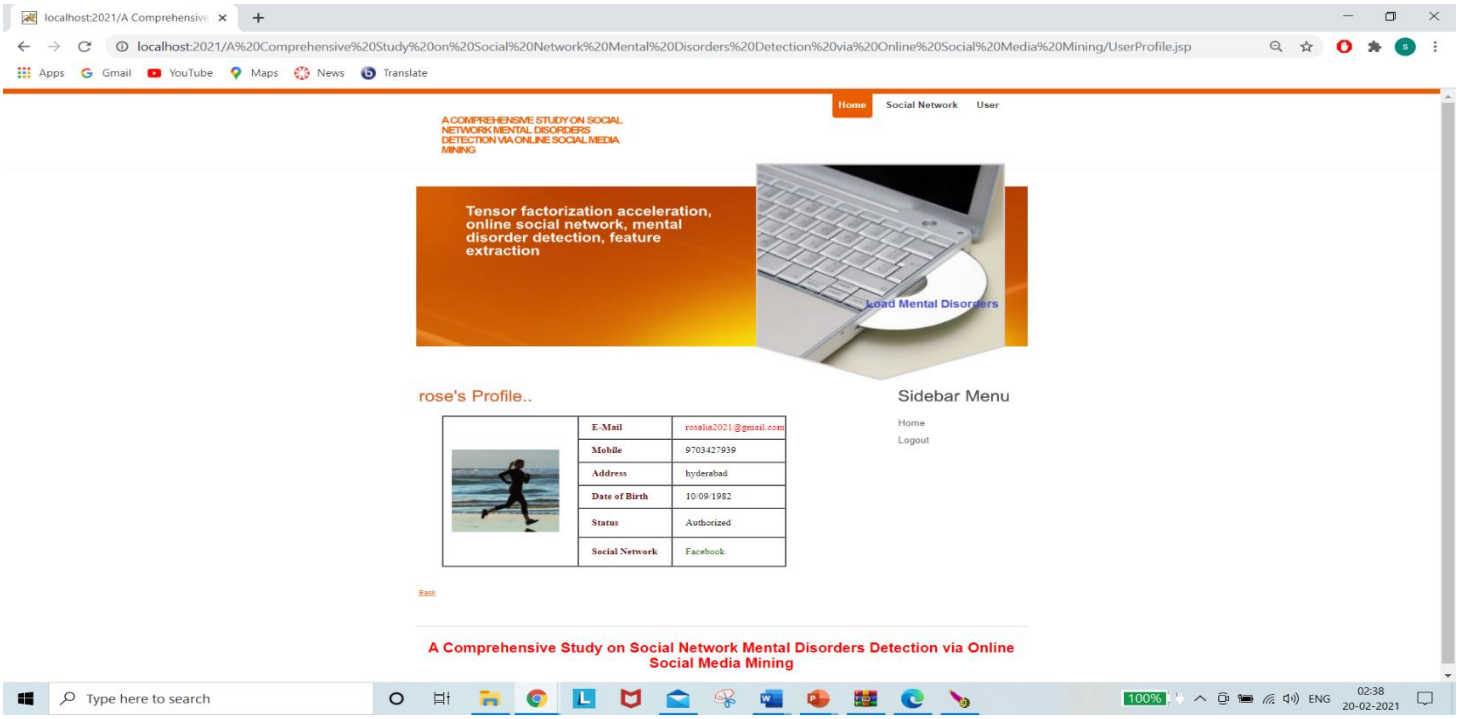


Fig. User profile

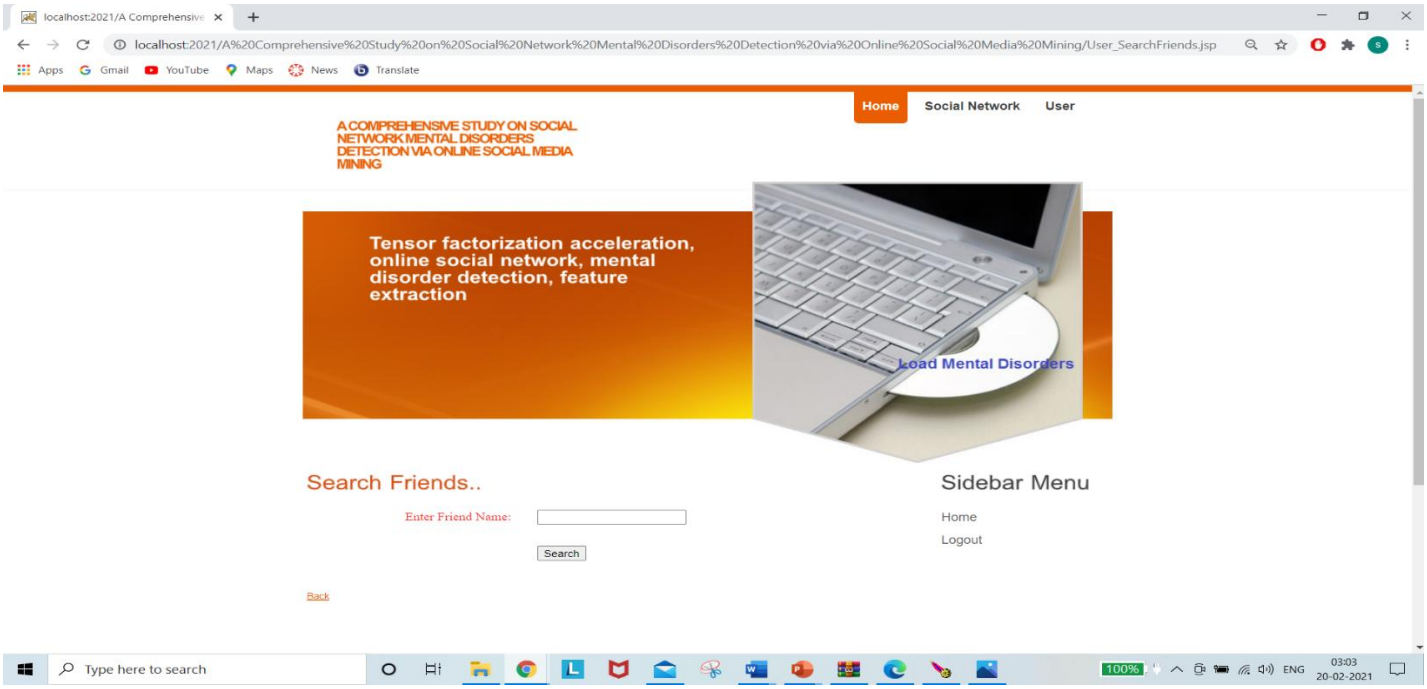


Fig. User can search friends

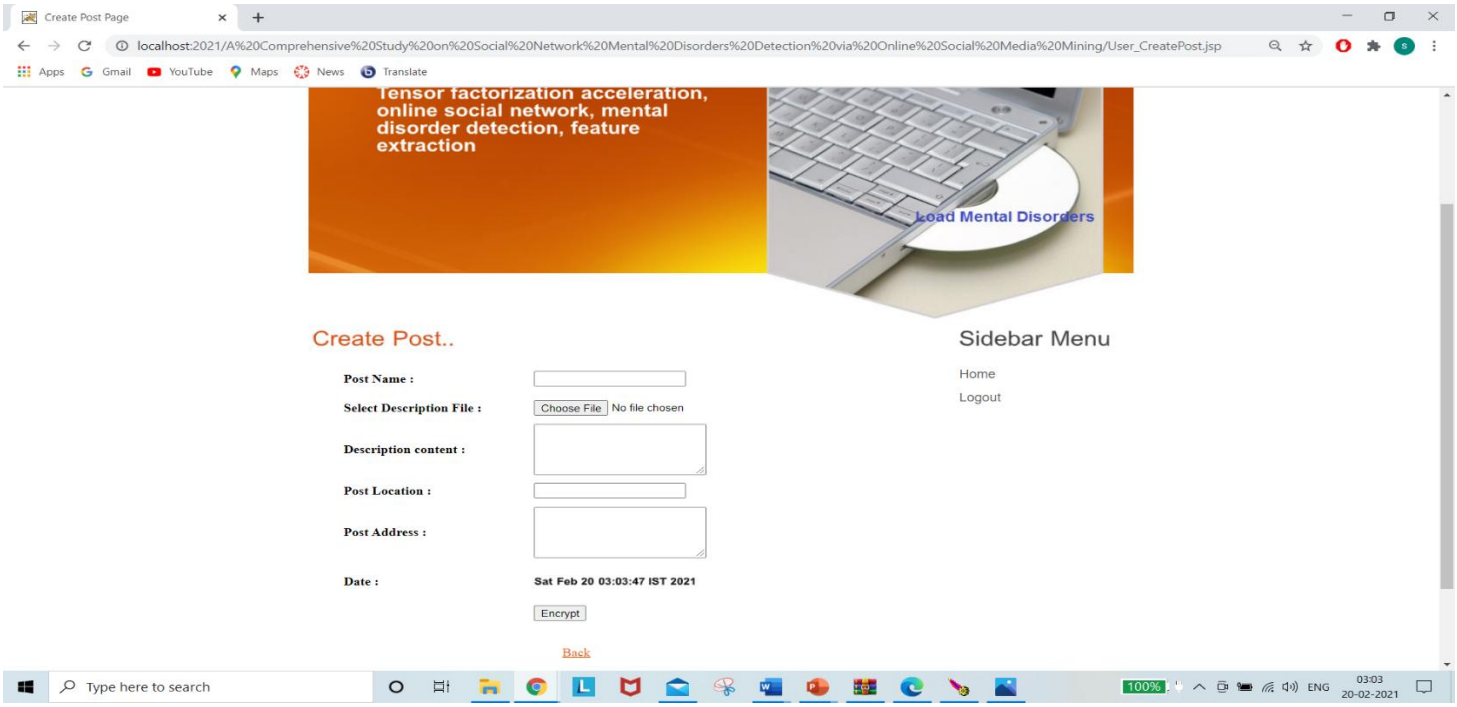


Fig. User can create a post

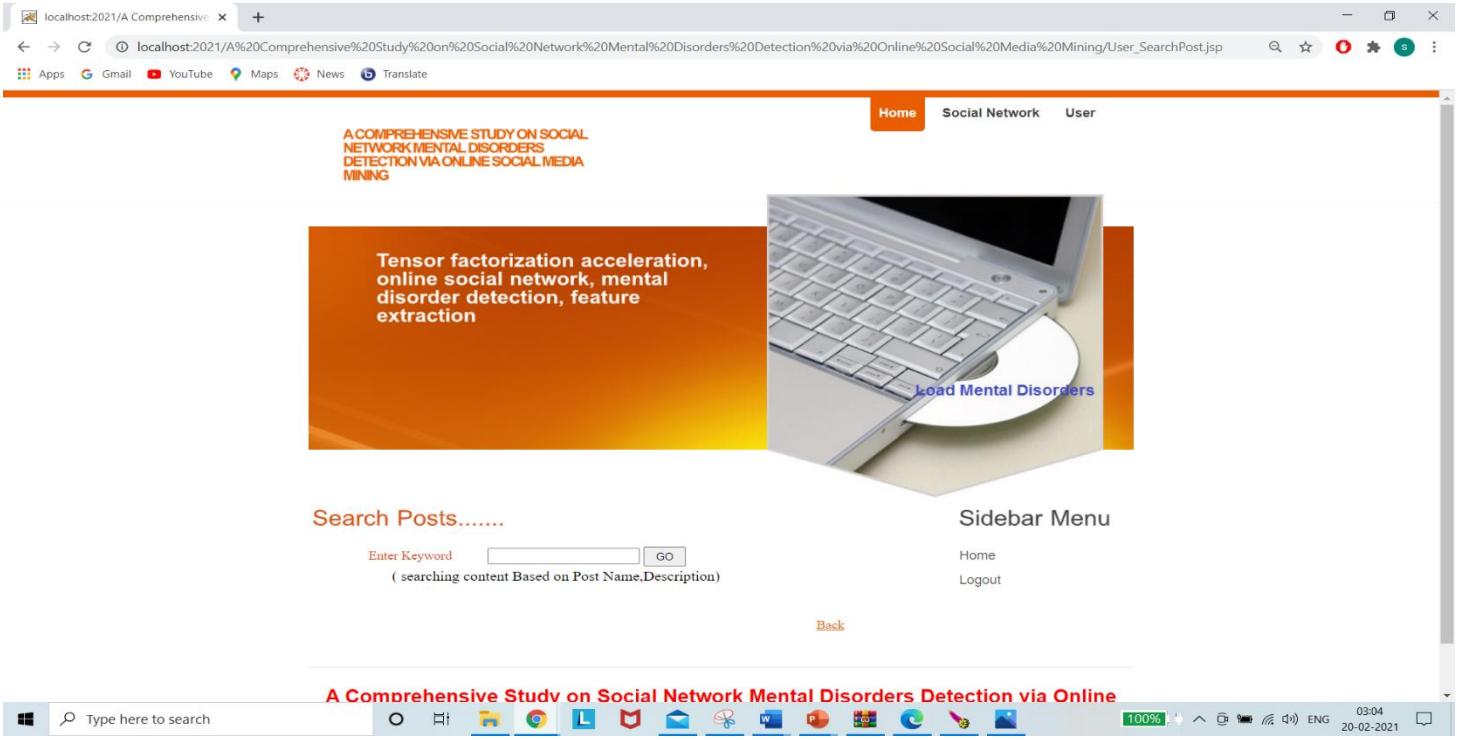


Fig. User can search posts and give comments and later logout

CHAPTER-9

CONCLUSION

In this paper, we make an attempt to automatically identify potential online users with SNMD's. We propose an SNMDD framework that explores various features from data logs of OSNs and a new tensor technique for deriving latent features from multiple OSNs for SNMD detection. This work represents a collaborative effort between computer scientists and mental healthcare researchers to address emerging issues in SNMDs. As for the next step, we plan to study the features extracted from multimedia contents by techniques on NLP and computer vision. We also plan to further explore new issues from the perspective of a social network service provider, e.g., Facebook or Instagram, to improve the well-beings of OSN users without compromising the user engagement.

CHAPTER-10

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